

## ZAKOPANE, Poland

WMO: 126250

Lat: 49.2939N

Lon: 19.9603E

Elev: 2812

StdP: 13.26

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 2.2        | 6.8 | -2.4                          | 5.4 | 3.8  | 2.3 | 6.9 | 8.6  | 15.4                  | 37.8 | 13.4 | 37.7 | 1.9                   | 170  | 0.542 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.2                   | 80.1            | 64.1 | 76.8 | 62.9 | 73.7 | 61.3 | 65.8                | 76.1 | 64.2 | 74.0 | 62.7 | 71.7 | 4.5                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 62.1                            | 92.5 | 70.0 | 60.5 | 87.5 | 68.5 | 59.1 | 83.0 | 66.2 | 32.2          | 76.8 | 30.9 | 74.0 | 29.7 | 71.8 | 72.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 12.4              | 10.2 | 8.7 | -3.5                       | 85.2 | 6.2                | 2.8 | -8.0                                               | 87.2 | -11.6      | 88.9 | -15.1      | 90.5 | -19.7      | 92.5 |    |    |
|                   |      |     | -4.1                       | 68.6 | 5.7                | 1.8 | -8.3                                               | 69.9 | -11.6      | 70.9 | -14.8      | 72.0 | -19.0      | 73.3 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 43.5   | 25.3  | 27.4  | 33.3  | 43.5  | 51.2  | 57.5  | 60.6  | 60.1  | 52.1  | 44.6  | 37.1  | 28.4 |
|                                                                                  | DBStd       | 14.69  | 9.10  | 9.39  | 8.51  | 7.69  | 6.77  | 6.39  | 5.69  | 5.80  | 6.52  | 8.01  | 8.96  | 9.11 |
|                                                                                  | HDD50       | 3539   | 767   | 633   | 518   | 219   | 67    | 10    | 1     | 3     | 52    | 205   | 395   | 669  |
|                                                                                  | HDD65       | 7902   | 1232  | 1053  | 983   | 645   | 428   | 236   | 158   | 174   | 388   | 633   | 838   | 1134 |
|                                                                                  | CDD50       | 1172   | 0     | 0     | 0     | 24    | 106   | 236   | 330   | 316   | 116   | 37    | 7     | 0    |
|                                                                                  | CDD65       | 58     | 0     | 0     | 0     | 0     | 1     | 12    | 22    | 21    | 2     | 0     | 0     | 0    |
|                                                                                  | CDH74       | 626    | 0     | 0     | 0     | 1     | 10    | 133   | 229   | 236   | 16    | 1     | 0     | 0    |
|                                                                                  | CDH80       | 90     | 0     | 0     | 0     | 0     | 0     | 15    | 33    | 40    | 2     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.3    | 3.4   | 3.7   | 4.0   | 3.8   | 3.5   | 3.0   | 2.9   | 2.7   | 2.6   | 3.2   | 3.6   | 3.5  |
| Precipitation                                                                    | PrecAvg     | 44.1   | 1.7   | 1.7   | 2.1   | 3.3   | 4.9   | 6.7   | 6.9   | 5.7   | 3.8   | 2.7   | 2.6   | 2.1  |
|                                                                                  | PrecMax     | 56.5   | 5.0   | 3.7   | 5.5   | 6.6   | 9.4   | 12.8  | 16.3  | 12.7  | 9.7   | 9.0   | 7.3   | 4.4  |
|                                                                                  | PrecMin     | 31.9   | 0.3   | 0.2   | 0.0   | 0.9   | 1.9   | 2.6   | 2.2   | 1.5   | 0.7   | 0.5   | 1.0   | 0.2  |
|                                                                                  | PrecStd     | 6.0    | 1.0   | 0.9   | 1.0   | 1.3   | 1.8   | 2.4   | 3.3   | 2.6   | 1.9   | 1.8   | 1.3   | 1.0  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 49.8  | 52.1  | 58.4  | 70.7  | 75.1  | 82.0  | 83.6  | 84.5  | 75.9  | 69.9  | 61.4  | 52.3 |
|                                                                                  |             | MCWB   | 40.7  | 40.6  | 44.8  | 53.5  | 58.5  | 65.4  | 65.2  | 64.7  | 60.6  | 56.0  | 49.5  | 44.7 |
|                                                                                  | 2%          | DB     | 45.6  | 47.4  | 54.0  | 66.0  | 71.7  | 78.0  | 79.9  | 79.8  | 71.7  | 65.3  | 57.3  | 48.3 |
|                                                                                  |             | MCWB   | 40.3  | 39.1  | 43.6  | 51.6  | 57.5  | 63.3  | 64.3  | 64.3  | 59.5  | 54.2  | 48.6  | 41.7 |
|                                                                                  | 5%          | DB     | 41.8  | 44.1  | 50.3  | 61.3  | 68.5  | 74.2  | 76.1  | 76.1  | 68.1  | 61.9  | 54.5  | 44.7 |
|                                                                                  |             | MCWB   | 37.7  | 38.2  | 41.8  | 48.8  | 56.1  | 61.5  | 63.1  | 63.0  | 58.0  | 52.7  | 47.0  | 39.4 |
|                                                                                  | 10%         | DB     | 37.7  | 40.4  | 46.4  | 57.2  | 64.8  | 70.5  | 72.8  | 72.5  | 64.8  | 58.2  | 51.1  | 41.4 |
|                                                                                  |             | MCWB   | 34.6  | 35.7  | 39.6  | 47.0  | 54.8  | 59.6  | 61.6  | 61.5  | 56.0  | 50.9  | 45.2  | 37.4 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 42.4  | 42.5  | 46.7  | 54.5  | 61.6  | 67.4  | 68.0  | 67.4  | 62.6  | 57.8  | 51.8  | 45.0 |
|                                                                                  |             | MCDB   | 46.6  | 48.8  | 54.6  | 67.8  | 71.0  | 78.0  | 78.2  | 78.1  | 72.3  | 66.2  | 58.1  | 50.7 |
|                                                                                  | 2%          | WB     | 40.8  | 40.5  | 44.6  | 52.2  | 59.2  | 64.5  | 65.9  | 65.4  | 60.6  | 55.4  | 49.5  | 42.9 |
|                                                                                  |             | MCDB   | 45.3  | 46.2  | 51.9  | 63.8  | 67.8  | 74.9  | 75.6  | 76.3  | 68.9  | 63.0  | 56.1  | 47.4 |
|                                                                                  | 5%          | WB     | 37.7  | 38.4  | 42.4  | 50.2  | 57.3  | 62.4  | 64.3  | 63.9  | 58.7  | 53.3  | 47.5  | 40.1 |
|                                                                                  |             | MCDB   | 41.3  | 43.4  | 49.1  | 59.6  | 65.7  | 71.6  | 73.3  | 74.1  | 66.3  | 60.3  | 53.6  | 43.9 |
|                                                                                  | 10%         | WB     | 35.0  | 36.1  | 40.3  | 47.9  | 55.3  | 60.6  | 62.7  | 62.2  | 56.9  | 51.3  | 45.4  | 37.5 |
|                                                                                  |             | MCDB   | 37.6  | 39.9  | 45.9  | 55.6  | 64.1  | 68.6  | 71.3  | 71.2  | 63.5  | 57.6  | 50.7  | 40.9 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 12.5  | 13.2  | 14.7  | 17.1  | 17.2  | 16.9  | 17.2  | 18.1  | 16.4  | 16.2  | 13.4  | 12.4 |
|                                                                                  |             | MCDBR  | 15.4  | 18.2  | 21.0  | 23.9  | 24.2  | 24.0  | 23.7  | 24.0  | 22.6  | 22.7  | 18.1  | 15.3 |
|                                                                                  |             | MCWBR  | 10.7  | 11.9  | 12.5  | 12.1  | 12.3  | 11.7  | 11.1  | 11.2  | 12.5  | 13.3  | 11.2  | 10.5 |
|                                                                                  | 5% WB       | MCDBR  | 14.4  | 16.2  | 18.9  | 21.4  | 21.7  | 21.2  | 20.8  | 22.0  | 20.1  | 21.5  | 16.3  | 14.3 |
|                                                                                  |             | MCWBR  | 10.4  | 11.1  | 11.9  | 11.7  | 11.9  | 11.1  | 11.0  | 11.3  | 11.7  | 13.2  | 10.6  | 10.5 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.286  | 0.313 | 0.360 | 0.420 | 0.416 | 0.418 | 0.431 | 0.425 | 0.392 | 0.355 | 0.323 | 0.284 |      |
|                                                                                  | taud        | 2.285  | 2.219 | 2.172 | 2.124 | 2.200 | 2.234 | 2.199 | 2.220 | 2.279 | 2.357 | 2.371 | 2.342 |      |
|                                                                                  | Ebn at Noon | 242    | 261   | 266   | 261   | 268   | 268   | 262   | 257   | 253   | 243   | 226   | 226   |      |
|                                                                                  | Edh at Noon | 23     | 31    | 38    | 45    | 43    | 43    | 43    | 41    | 35    | 27    | 21    | 19    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 288    | 526   | 886   | 1290  | 1550  | 1657  | 1606  | 1471  | 1037  | 633   | 323   | 223   |      |
|                                                                                  | RadStd      | 44     | 79    | 132   | 148   | 198   | 156   | 183   | 131   | 152   | 90    | 44    | 25    |      |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |      | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |      | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j)  | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A  | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (46 neighbors) | +1.03 | N/A     | N/A    | +1.71 | +0.90   | +0.77 | -234 | -331        | +158  | +68   |       |

Nomenclature: See separate page